MyKad OCR & Vision Research Complete Physical & Technical Reference

Kesh (deadboy18)

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1. Executive Summary

This document is the definitive technical reference for the Malaysian National Identity Card (MyKad) system, covering physical specifications, security features, manufacturing process, chip architecture, IC number format, and OCR/Vision feasibility analysis. It is written for developers from any country who need to understand how the MyKad works at every level from the polycarbonate layers and laser engraving to the APDU commands that read data from the smart card chip.

Bottom Line on OCR: Free OCR engines (Tesseract, PaddleOCR, EasyOCR) cannot reliably read MyKad text. Vision LLM APIs (GPT-4.1-nano) achieve 98–100% accuracy. USB chip readers achieve 100% accuracy offline.

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2. Card Overview & History

2.1 What is MyKad

MyKad (Malaysia Kad) is the compulsory national identity card issued by Jabatan Pendaftaran Negara (JPN National Registration Department) to all Malaysian citizens aged 12 and above. Introduced on 5 September 2001, it replaced the High Quality Identity Card (Kad Pengenalan Bermutu Tinggi).

Malaysia was the first country in the world to use an identification card incorporating both photo identification and fingerprint biometric data on an embedded computer chip. The technology was developed by IRIS Corporation Berhad, a Malaysian company that also invented the world's first ePassport in 1998.

Sources: - Wikipedia: Malaysian identity card https://en.wikipedia.org/wiki/Malaysian_identity_card - IRIS Corporation Berhad https://www.iris.com.my/ - Malaysia Government Portal https://www.malaysia.gov.my/en/categories/personal-identification/identification-card

MyKad Evolution

Figure 4: Evolusi Kad Pengenalan Evolution of Malaysian Identity Cards (Source: JPN Official)

2.2 Evolution of Malaysian Identity Cards

The Malaysian identity card system has evolved through multiple generations:

Year	Card Name	Region	Notes
1948	Kad Pengenalan (Kertas)	Tanah Melayu (Peninsular)	Paper card, known as the "Rice Card", issued to curb communist threat
1949	Kad Pengenalan (Kertas Berlipat)	Sarawak	Folded paper card
1960	Kad Pengenalan (Plastik)	Semenanjung Malaysia	First plastic card for Peninsular Malaysia
1960	Kad Pengenalan (Bersalut)	Sarawak	Coated card
1963–1972	Kad Pengenalan (Oren)	Sabah	Orange card, 6-digit number
1966	Kad Pengenalan (Plastik)	Sarawak	Plastic card
1972	Kad Pengenalan Plastik (Biru)	Sabah	Blue plastic card, 7-digit number
1990	Kad Pengenalan	All Malaysia	High Quality Identity Card, introduced 12-digit IC format

	Bermutu Tinggi		(YYMMDD-PB-####)
1999	Kad Pengenalan Bermutu Tinggi (Updated)	All Malaysia	Updated security features
2001	MyKad (Kad Pengenalan Pelbagaiguna Kerajaan)	All Malaysia	Government Multi-Purpose Card, 32KB chip, contact-only
2002	MyKad (64KB)	All Malaysia	Chip upgraded to 64KB, thumbprint data added
2012	MyKad Struktur Baru	All Malaysia	New structure, 80KB chip, hybrid contact+contactless (TnG)
2026 (planned)	Next-Gen MyKad	All Malaysia	QR code feature, enhanced security, polycarbonate with laser engraving and hologram

Source: JPN Official Evolusi Kad Pengenalan infographic (Jabatan Pendaftaran Negara, Kementerian Dalam Negeri) **Source:** Lowyat.NET https://www.lowyat.net/2025/374652/malaysia-next-gen-mykad-qr-code-feature/ **Source:** iLoveBorneo https://www.iloveborneo.my/mykad-lama-bakal-digantikan-ketahui-ciri-ciri-kad-pengenalan-baharu/

2.3 Card Variants by Colour

Card	Issued To	Colour	Hex (Approx.)	Malay Label
MyKad	Citizens aged 12+	Blue	#0032A0	KAD PENGENALAN MALAYSIA
MyKid	Citizens under 12	Pink	#E91E63	KAD PENGENALAN MALAYSIA (KANAK-KANAK)
MyPR	Permanent residents	Red	#CC0000	KAD PENGENALAN MALAYSIA PEMASTAUTIN TETAP
MyKAS	Temporary residents	Green	#008000	KAD PENGENALAN SEMENTARA
MyTentera	Military personnel	Silver	#C0C0C0	Armed Forces logo, replacing BAT C 10 form
MyPolis	Police personnel	Varies		Police variant

Source: ClearTax Malaysia https://www.cleartax.com/my/en/mykad-malaysia

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3. Physical Card Specifications

3.1 Dimensions & Material

Property	Value
Card Standard	ID-1 (ISO/IEC 7810) same as a standard credit card
Width	85.6 mm (3.370 inches)
Height	53.98 mm (2.125 inches)
Thickness	0.76 mm (0.030 inches)
Corner Radius	3.18 mm (standard ID-1)
Material	Polycarbonate (PC) NOT PVC
Weight	~5–6 grams
Primary Colour	MyKad Blue (#0032A0 approximate)
Orientation	Landscape (wider than tall)
Lifespan	10 years (card body), 20 years (chip data retention)
Testing Standard	ISO 10373

Source: CardLogix https://www.cardlogix.com/product/id-grade-polycarbonate-card-cr80-30-mil-laser-engravable/ **Source:** Datasonic/APCS https://www.apcscard.com/polycarbonate-card-laser-engrave-coated-overlay/

3.2 Why Polycarbonate, Not PVC

Polycarbonate is chosen over PVC for several critical reasons: it is the same material used for bulletproof glass and CDs. Multiple polycarbonate layers fuse together under heat and pressure into a "monolithic block" the layers cannot be separated without destroying the card. This makes photo swapping or data tampering physically impossible without visible damage. PVC cards (like credit cards) use adhesive lamination which can be peeled apart.

Source: DSS Plastics / PolycarbonateCard.com https://www.polycarbonatecard.com/more/

3.3 Embedded Chips

Contact Chip (Identity Data)

Property	Value
Standard	ISO 7816
Type	Contact smart card
Location	BACK of card, lower-right quadrant
Visible As	Gold contact pad with 8 pins
EEPROM	80KB (current), was 32KB (2001), then 64KB (2002)
Operating	M-COS (MyKad Chip Operating System), proprietary to IRIS Corporation

System	
Access	Reading: no PIN required. Writing: requires JPN Perso Card + Triple-DES key

Contact Chip Data Contents

Data Field	Approximate Size	Notes
IC Number	12 bytes	Stored as ASCII digits, no dashes
Full Name	Up to 80 bytes	UTF-8 encoded
Date of Birth	8 bytes	YYYYMMDD format
Gender	1 byte	M or F
Race	Variable	Malay encoding
Religion	Variable	Malay encoding
Nationality	Variable	WARGANEGARA or PENDUDUK TETAP
Address	~200 bytes	3 lines + postcode + city + state
JPEG Photo	~15KB	Stored in 4000-byte allocation, ~200×250 pixels
Fingerprint (Right Thumb)	~500 bytes	IRIS Corp proprietary format (NOT NIST/ISO minutiae)
Fingerprint (Left Thumb)	~500 bytes	Same proprietary format
SOCSO/PERKESO Number	Variable	Social security reference
JPJ Driving Licence	Separate application	On same chip, different file
Immigration Data	Separate application	Passport references
PKI Digital Certificates	Optional	Can store 2 certificates from MSCTrustgate or DigiCert

Contactless Chip (Touch 'n Go)

Property	Value
Standard	ISO 14443-3A
Type	Mifare Classic 1K
Location	Embedded inside card body (not visible)
Sectors	16 sectors, each with proprietary sector keys
Access	Keys held by Touch 'n Go Sdn Bhd unauthorised readers cannot access
Data	E-wallet stored value and transaction history ONLY no personal identity data

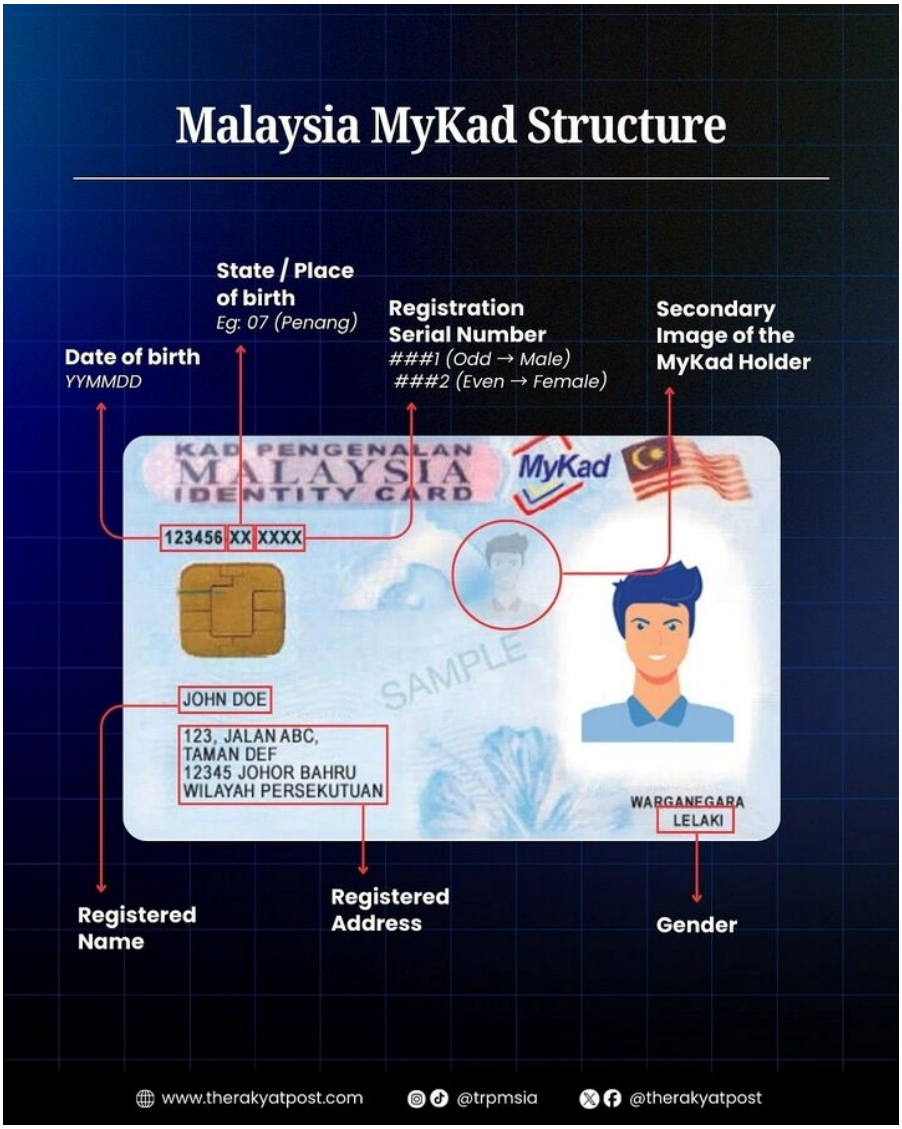
NFC Readable	Detectable by NFC readers (ACR122U etc.) but data inaccessible without keys

Can the Chip Data Be Modified?

Contact chip: NO not by the public. Reading requires only a PC/SC reader and standard APDU commands (documented in MyKad_Complete_Technical_Reference_v3_LLM.md). Writing requires: (1) a JPN Personalisation Card (a special operator smart card), (2) Triple-DES authentication keys stored in Hardware Security Modules (HSMs) at JPN/IRIS Corporation facilities. Without both, all write commands are rejected by the chip's access control system.

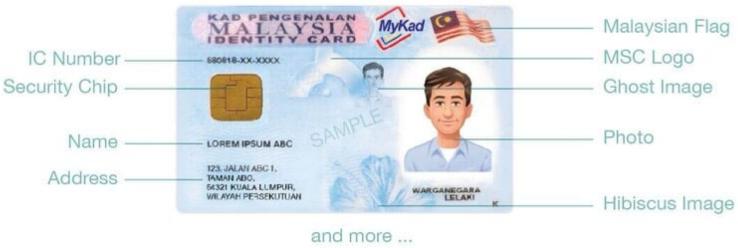
Contactless TnG chip: Only by authorised Touch 'n Go terminals. The Mifare Classic 1K sector keys are proprietary. While Mifare Classic has known cryptographic weaknesses (Crypto-1 has been broken by academic researchers), exploiting this on a production MyKad would be illegal under Malaysian law.

Source: IRIS Corporation https://www.iris.com.my/smart-card-solution/ **Source:** Innov8tif https://innov8tif.com/3-methods-to-read-mykad-2/



MyKad Structure Diagram

Figure 1: Malaysia MyKad Structure (Source: The Rakyat Post)



MyKad Front Layout

Figure 2: MyKad Front Face Element Map

4. Front Face Detailed Element Map

All positions are described as percentages of total card width (W=85.6mm) and height (H=53.98mm), measured from the top-left corner. This mapping is essential for OCR region detection and YOLO training.

4.1 Header Region (Top 0–18% H)

Element	Position	Details
"KAD PENGENALAN"	Top-left, 5–40% W, 0–6% H	White/silver text, small font, serif-like
"MALAYSIA"	Below KAD PENGENALAN, 5–40% W, 5–12% H	LARGEST header text, white/silver, all caps
"IDENTITY CARD"	Below MALAYSIA, 5–40% W, 10–15% H	Smaller English text
Jata Negara (Coat of Arms)	Centre, ~42–55% W, 0–15% H	Small printed emblem, ~8mm wide
"MyKad" Logo	Right of centre, ~58–75% W, 2–12% H	Stylised script font with MSC Malaysia branding
Malaysian Flag	Top-right, ~78–95% W, 2–12% H	Jalur Gemilang, small printed flag

4.2 Personal Data Region (18–95% H)

Element	Position	Details
IC Number	18–32% H, 15–65% W	LARGEST TEXT ON CARD. Format: YYMMDD-PB-####. Laser-engraved monospaced font. This is the primary OCR target and the hardest to read due to laser engraving over guilloche.
Contact Chip Pad	25–65% H, 3–22% W	Gold ISO 7816 contact pad. 8-pin layout. Positioned on the LEFT side of the card front.
Ghost Image (Secondary)	25–65% H, ~45–65% W	Small, translucent laser-engraved portrait of the holder. Semi-transparent, overlaid on the guilloche pattern. Serves as anti-tampering feature must match the main photo.
Main Photo	20–80% H, 65–95% W	Colour photograph, ~25×32mm. Blue background. Positioned on the RIGHT side. Printed under polycarbonate overlay. No guilloche behind the photo area.
Full Name	55–68% H, 3–55% W	ALL UPPERCASE. Laser-engraved. May include BIN/BINTI (Malay), A/L/A/P (Indian), or @ (alias).

Address	68–90% H, 3–55% W	1–3 lines, printed (not engraved), smaller font. Includes street, postcode (5 digits), city, state. ALL UPPERCASE.
Nationality	85–95% H, 50–75% W	"WARGANEGARA" (Citizen) or "PENDUDUK TETAP" (PR)
Religion	Below nationality	ISLAM / KRISTIAN / BUDDHA / HINDU / SIKH / TIADA AGAMA
Gender	85–95% H, 75–95% W	LELAKI (Male) / PEREMPUAN (Female)
Hibiscus Watermark	Bottom-right corner, ~90–98% W, 85–98% H	Small "K" or hibiscus flower symbol (Bunga Raya national flower)

4.3 Photo Specifications (JPN Requirements)

Property	Value
Size on card	~25mm × 32mm
Background	Blue (#0066CC approximate)
Face position	Centred, 70–80% of frame height
Resolution on chip	~200 × 250 pixels, JPEG, ~15KB
Taken at	JPN counter with standardised camera setup

Source: The Rakyat Post MyKad Structure infographic (www.therakyatpost.com) **Source:** CTOS eKYC https://ctoscredit.com.my/business/evaluate-new-customer/ctos-ekyc/document-id-holder-verification/

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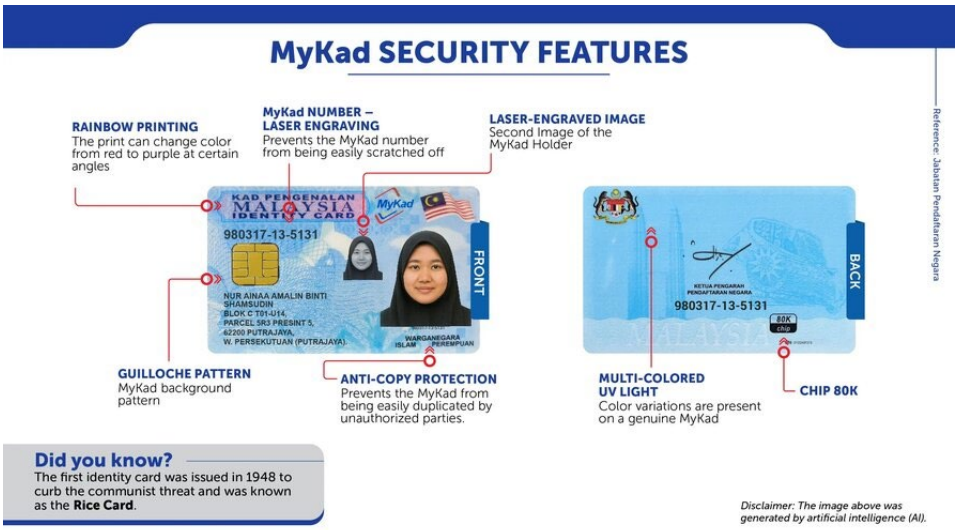
5. Back Face Element Map

Element	Position	Details
Ghost Image (Large)	Upper-centre, 5–45% H	Laser-engraved monochrome portrait, larger than front ghost
"KETUA PENGARAH PENDAFTARAN NEGARA"	Centre, 45–55% H	Director General title with printed signature (not personalised)
IC Number (Repeated)	Centre, 55–65% H	Same YYMMDD-PB-#### format, printed (NOT laser-engraved on back), easier to read optically
80K Chip Pad	Lower-right, 65–90% H, 75–95% W	Gold ISO 7816 contact pad, labelled "80K chip"
Serial Number	Bottom edge, 90–	Small printed manufacturing serial (e.g.,

	98% H	“D1024NF513”), NOT the IC number
Card Count	After IC number	2-digit suffix indicating how many MyKads the person has held



MyKad Security Features

Figure 3: MyKad Security Features Overview (Reference: Jabatan Pendaftaran Negara)

6. Security Features Technical Analysis

6.1 Laser Engraving

- **Location:** IC number, name, ghost image (front), ghost image (back)
- **Technology:** Ytterbium fibre-doped laser, up to 1200 DPI resolution
- **Process:** Laser beam causes carbonisation of the polycarbonate polymer at the focal point, creating darkened marks INSIDE the card body (not on the surface)
- **OCR Impact:** CRITICAL variable reflectivity depending on viewing angle and lighting direction. The same digit appears differently in every photo.
- **Source:** IAI Industrial Systems https://www.iai.nl/personalisation-experts/personalisation-technologies/laser-engraving/
- **Source:** CampusIDNews https://www.campusidnews.com/laser-engraving-ids-catching-on/
- **Source:** Identis Group https://identisgroup.com/newsroom/2018/03/laser-personalization-for-high-security-id-cards/

6.2 Guilloche Pattern

- **Location:** Entire card surface (both sides), except photo area
- **Description:** Mathematically-generated interlocking curved lines (hypotrochoid/spirograph pattern)

- **Line width:** ~0.1mm, visible under magnification
- **Generation:** Requires specialised security design software with complex mathematical parameters. The exact parameters are secret.
- **OCR Impact:** SEVERE OCR engines interpret fine lines as text fragments
- **Source:** Ben Hodosi (Security Document Designer) https://benhodosi.com/
- **Source:** SZIMAGETECH https://szimagetech.com/explore-comprehensive-id-card-security-features/

6.3 Rainbow Printing (Iris/Split-Fountain)

- **Location:** Card background, most visible at edges
- **Process:** Offset printing with two colours loaded in the same ink fountain with gradient blend
- **Visual effect:** Colour shifts from red to purple when card is tilted
- **OCR Impact:** MODERATE breaks colour-based binarisation

6.4 Holographic Overlay (DOVID)

- **Location:** Transparent overlay across front face
- **Type:** Diffractive Optically Variable Image Device (DOVID)
- **Structure:** Nano-scale diffractive gratings embossed into thin film, metallised with aluminium via vacuum deposition
- **Visual effects:** Rainbow diffractive colours, dynamic images that change with viewing angle
- **OCR Impact:** SEVERE creates moving specular reflections (glare)
- **Source:** OVD Kinegram https://www.kinegram.com/events-insights/details/what-is-secure-hologram-technology
- **Source:** Wikipedia (DOVID) https://en.wikipedia.org/wiki/Diffractive_optically_variable_image_device

6.5 UV Fluorescent Printing

- **Location:** Various positions across card
- **Technology:** Special fluorescent compounds in ink that react to ultraviolet (365nm) light
- **Visible under:** UV/black light only invisible under normal lighting
- **OCR Impact:** None under normal lighting
- **Source:** Troy Group https://www.troygroup.com/blog/uv-ink-for-security

6.6 Anti-Copy Protection

- **Features:** Micro-printing (<0.5mm characters), moiré interference patterns, fine-line structures
- **Effect:** Photocopies and scans show visible degradation artifacts
- **Source:** ID Management https://www.idmanagement.com/card-security

6.7 Hibiscus Watermark

- **Location:** Bottom-right corner of front face
- **Description:** Small printed Bunga Raya (hibiscus Malaysian national flower) symbol or letter "K"

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7. Manufacturing Process Step by Step

7.1 Card Body Production (Factory IRIS Corporation / Datasonic)

Step 1: Polycarbonate Sheet Preparation Multiple layers of polycarbonate film (~0.076mm per layer) are prepared. Each layer serves a specific purpose: core layers for structural integrity, print-receiving layers for offset artwork, and laser-sensitive layers containing polymer that carbonises when hit by a laser beam.

Step 2: Offset Printing The guilloche pattern, rainbow gradient, UV fluorescent elements, fixed text ("KAD PENGENALAN MALAYSIA / IDENTITY CARD"), MyKad logo, Malaysian flag, and coat of arms are printed onto inner polycarbonate layers using offset lithography at 2400+ DPI. This is done in bulk thousands of card sheets at once.

Step 3: Chip Embedding The ISO 7816 contact chip module (80KB EEPROM with M-COS operating system) and the Mifare Classic 1K contactless antenna/chip are placed between polycarbonate layers at precise positions. The contact chip goes to the front-left position, the contactless antenna is embedded throughout the card body.

Step 4: Lamination (Fusion) All polycarbonate layers are fused together under high heat (~180°C) and pressure in an industrial lamination press. No adhesives are used the polycarbonate layers molecularly bond into a single monolithic block. This is the key security step: the layers CANNOT be separated without destroying the card.

Step 5: Die Cutting Individual cards are precision-cut from the laminated sheet to ID-1 dimensions (85.6 × 53.98mm) with 3.18mm corner radius.

Step 6: DOVID Application The holographic overlay (DOVID) is heat-sealed or transferred onto the card surface. This is a thin film (<10 microns) containing nano-scale diffractive structures.

Result: A blank card body with all security printing, chips, and hologram ready for personalisation.

7.2 Personalisation (At JPN Counter or Regional Centre)

Step 7: Enrolment At the JPN counter, the officer captures: digital photo (against blue background), both thumbprints (via biometric scanner), and verifies/enters personal data into the JPN system.

Step 8: Laser Personalisation (~30–60 seconds) The blank card body is loaded into a laser engraving machine. A Ytterbium fibre laser engraves: IC number (front, large), full name (front), passport photo (front, greyscale laser image), ghost image (front, semi-transparent), and ghost image (back, monochrome). The laser operates at different power levels to produce 256 greyscales for the photographic images.

Step 9: Chip Personalisation Via APDU commands authenticated by the JPN Perso Card + Triple-DES keys: personal identity data, JPEG photo (~15KB), fingerprint templates, residential address, and SOCSO number are written to the contact chip. The TnG contactless chip is initialised if applicable.

Step 10: Quality Control Automated optical inspection verifies laser engraving quality. Chip read-back verifies all data was written correctly. Card is tested against ISO 10373 standards.

Step 11: Delivery Card is handed to the applicant at the counter (for walk-in applications) or mailed to the registered address.

Equipment at JPN: - Laser engraving system (Ytterbium fibre laser, ~$50,000–$100,000+) - Contact smart card reader/writer with Perso Card slot - Biometric fingerprint scanner (FBI-certified) - Digital camera with blue background - JPN personalisation software connected to central database

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8. What It Would Take to Counterfeit a MyKad

This section is included for security awareness understanding the barriers helps developers appreciate why the card's physical security is effective.

Component	Equipment Needed	Approximate Cost	Availability
Polycarbonate card stock (ID-grade, multi-layer)	Specialist supplier	RM50–200 per sheet	Controlled tracked by suppliers
Offset printing (guilloche, UV, rainbow)	Offset lithography press, 2400+ DPI	RM500,000+	Available but requires guilloche design codes
Chip module (80KB EEPROM, M-COS)	IRIS Corporation proprietary	NOT AVAILABLE	M-COS is proprietary, not sold publicly
Mifare Classic 1K	Commercial supplier	RM2–5 per chip	Available, but sector keys are TnG proprietary
Lamination press	Industrial	RM1,000,000+	Available from

(polycarbonate fusion)	lamination press, 180°C		industrial suppliers
Holographic overlay (DOVID)	Hologram origination + replication	RM50,000+ (origination) + RM1+/card	10–12 weeks lead time, custom design
Laser engraving system	Ytterbium fibre laser, 1200 DPI	RM200,000–450,000	Available but controlled
Chip personalisation	JPN Perso Card + Triple-DES keys	NOT AVAILABLE	Keys stored in HSMs at JPN/IRIS only
Total minimum investment	All above	RM2,000,000+	Multiple items are NOT publicly available

Key barriers: 1. The M-COS chip operating system is proprietary to IRIS Corporation you cannot buy it 2. The JPN Perso Card and Triple-DES keys are held exclusively by JPN you cannot obtain them 3. Even with all equipment, the guilloche mathematical parameters are secret 4. Any card reader running official IRIS SDK will detect a counterfeit chip immediately 5. Attempting to counterfeit a MyKad is a serious criminal offence under Malaysian law

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9. Next-Generation MyKad (2026)

Malaysia's Deputy Home Minister, Datuk Seri Dr. Shamsul Anuar Nasarah, announced that new MyKad cards will begin issuance from June 2026. Key changes:

- **QR Code:** A dedicated QR code for digital on-the-spot authentication by authorities
- **Enhanced laser engraving and hologram** features
- **Polycarbonate construction** (continuing from current generation)
- **Larger chip capacity** and higher security standards
- **Biometric data expansion** potentially additional biometric modalities beyond thumbprints
- All functions of the old MyKad will be discontinued once the new generation is fully rolled out

The new supply contract also covers updated versions of MyTentera (Armed Forces) and MyPoCA (ex-convict) cards.

Source: Lowyat.NET (27 Nov 2025) https://www.lowyat.net/2025/374652/malaysia-next-gen-mykad-qr-code-feature/ **Source:** iLoveBorneo (20 Nov 2024) https://www.iloveborneo.my/mykad-lama-bakal-digantikan-ketahui-ciri-ciri-kad-pengenalan-baharu/

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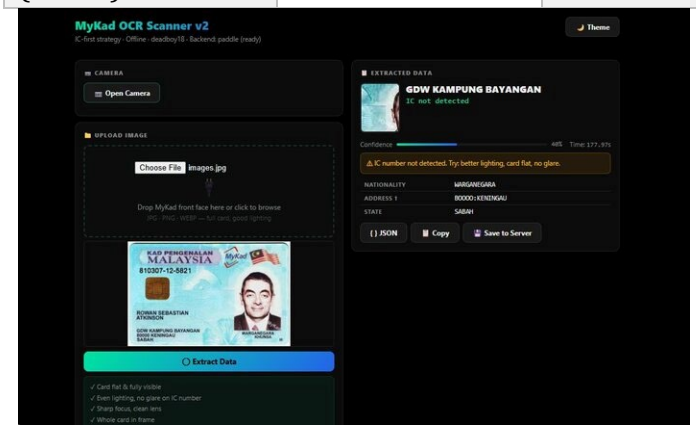
10. OCR & Vision Feasibility Test Results

10.1 Approaches Tested

Approach	IC Accuracy	Name Accuracy	Speed (CPU)	Verdict
Tesseract 5.x	30–40%	20–30%	~2s/region	NOT VIABLE
PaddleOCR v5	50–60%	40–50%	~300s total pipeline	NOT VIABLE
EasyOCR	~50–60%	~40–50%	Similar to PaddleOCR	NOT VIABLE
YOLO v8-nano + PaddleOCR	55–65%	45–55%	~180s	NOT VIABLE
Multi-Pass OCR (5 variants)	+5–10% improvement	Marginal	~300s (20+ OCR calls)	NOT VIABLE too slow
Checksum Error Correction	+5–15% on IC	N/A	<1ms	Helps but insufficient alone
Vision LLM (GPT-4.1-nano)	98–100%	95–100%	2–3s	WORKS requires internet
USB Chip Reader (APDU)	100%	100%	3–5s	GOLD STANDARD fully offline



OCR Test Result

Figure 5: OCR Pipeline Test Result IC number not detected (mykad_ocr_v2.py with PaddleOCR + YOLO)

10.2 Why Vision LLMs Succeed Where OCR Fails

Vision LLMs don't read characters independently they apply contextual reasoning. The LLM knows an IC number must be YYMMDD-PB-####, validates the date, checks the state code, and cross-references the gender digit parity. OCR engines treat each character independently with no structural understanding.

Critical note: Vision LLM accuracy is highest with a FIXED camera position relative to the card. For kiosk deployments, use a fixed-mount camera with a card slot. Handheld phone photos achieve ~90–95% accuracy (vs 98–100% fixed).

10.3 Recommended Architecture

1. **Primary:** USB smart card chip reader (offline, 100% accurate, 3–5 seconds)
2. **Secondary:** Vision LLM API for photo-based flows (internet required, ~$0.001/scan)
3. **Tertiary:** OCR pipeline as degraded fallback (low accuracy, flag for manual verification)

10.4 The Pre-processing "Graveyard" (What Doesn't Work)

Developers often assume that applying standard OpenCV filters will magically make the laser-engraved text readable. Extensive testing confirms that software pre-processing cannot fix the optical interference caused by the guilloche pattern.

The following standard approaches were tested and failed:

- **Adaptive Thresholding:** Fails because the dark lines of the guilloche pattern intersect with the laser-engraved text, causing the characters to bridge together into unreadable blobs.
- **CLAHE (Contrast Limited Adaptive Histogram Equalization):** Amplifies the background noise of the rainbow printing just as much as the text it's supposed to enhance.
- **Canny Edge Detection:** Creates a chaotic web of edges due to the holographic DOVID overlay the diffractive nanostructures produce edge responses indistinguishable from actual text strokes.
- **Morphological Dilation/Erosion:** Smudges the tightly packed 12-digit IC number into an unreadable solid block. The digit spacing is too tight relative to the guilloche line spacing for any structuring element to separate them cleanly.
- **Morphological Black-Hat (Guilloche Removal):** Produces visually cleaner digit images but PaddleOCR still misreads them frequently. The laser engraving's variable reflectivity defeats any fixed-threshold approach.
- **Multi-Pass OCR (5 Preprocessing Variants):** Running OCR on 5 different preprocessed versions (raw, adaptive threshold, Otsu, CLAHE, sharpened) gives marginal accuracy improvement (+5–10%) at catastrophic speed cost (20+ OCR calls × 15s each = 300s total).

Conclusion: Do not waste time trying to clean the image via OpenCV. Rely on contextual AI (Vision LLMs) or skip optics entirely via the smart card chip.

10.5 Hardware Mitigation: Cross-Polarised Lighting

If you are building a physical kiosk and must rely on optical scanning, you can physically eliminate the severe glare from the Holographic Overlay (DOVID) using cross-polarisation.

- **The Setup:** Place a linear polarising film over your kiosk's LED light source. Place a second polarising film over the camera lens, rotated exactly 90 degrees offset from the light's filter.
- **The Result:** This completely blocks all direct specular reflections (glare) bouncing off the plastic and hologram, allowing the camera to see straight through to the printed and laser-engraved layers beneath the DOVID.
- **Cost:** Polarising film sheets are approximately RM20–50 per sheet. This is the cheapest hardware modification that yields the most significant optical improvement.
- **Limitation:** Cross-polarisation eliminates glare but does NOT eliminate the guilloche pattern interference, which is a printed feature inside the card body. OCR will still struggle with the guilloche even with perfect glare-free images but Vision LLMs handle it significantly better when glare is removed.

10.6 UI/UX Fallback Strategies

Because optical extraction (even via Vision LLMs) has a 5–10% failure rate when users upload handheld photos, your software must handle failures gracefully.

- **Do not block the user:** If the LLM returns low confidence or if the extracted IC number fails the Mod 11,2 checksum validation, do not throw a hard error. Dead-end error screens frustrate users and increase abandonment rates.
- **Pre-fill and Flag:** Pre-fill the form with whatever data was successfully extracted, highlight the failing fields (e.g., in yellow), and prompt the user to manually correct only the flagged values. This turns a failed scan into a 5-second manual correction instead of a full re-entry.
- **Auto-retry with guidance:** If the first scan fails, show the user specific guidance: "Hold the card flat, avoid shadows on the IC number area, ensure even lighting." Then allow a one-tap re-scan rather than requiring re-upload.
- **Checksum as silent validator:** Run the ISO 7064 Mod 11,2 checksum on the extracted IC number in the background. If it passes, the IC is almost certainly correct. If it fails, flag the IC field specifically and tell the user which digits look suspicious (based on the checksum correction algorithm's best guesses).

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11. Technical Assets & Files

File	Description
mykad_ocr_v2.py (~1,700 lines)	Complete OCR pipeline: Flask web UI, webcam, upload, YOLO, PaddleOCR, checksum correction
train_yolo.py (~180 lines)	YOLOv8-nano training script for 6-class MyKad field detection
mykad_ic_best.pt	Trained YOLOv8-nano weights

Scan_Image_Service.go (~130 lines)	Go service for Vision LLM API scanning
Scan_Image_Base.go (~50 lines)	Base prompt for Vision LLM extraction
MyKad_Complete_Technical_Reference_v3_LLM.md (~1,400 lines)	23-chapter technical reference
Self_Validating_12_Digit_Identifier_System.md (~300 lines)	Abstracted checksum specification

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Kesh (deadboy18) April 2026 Confidential Research Document Addendum to: MyKad Developer Ecosystem Research Portfolio